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COURSE: COMPUTER SYSTEMS ENGINEERING

SUBJECT: EMS316D PROJECT REPORT

DATE: 13 May 2026

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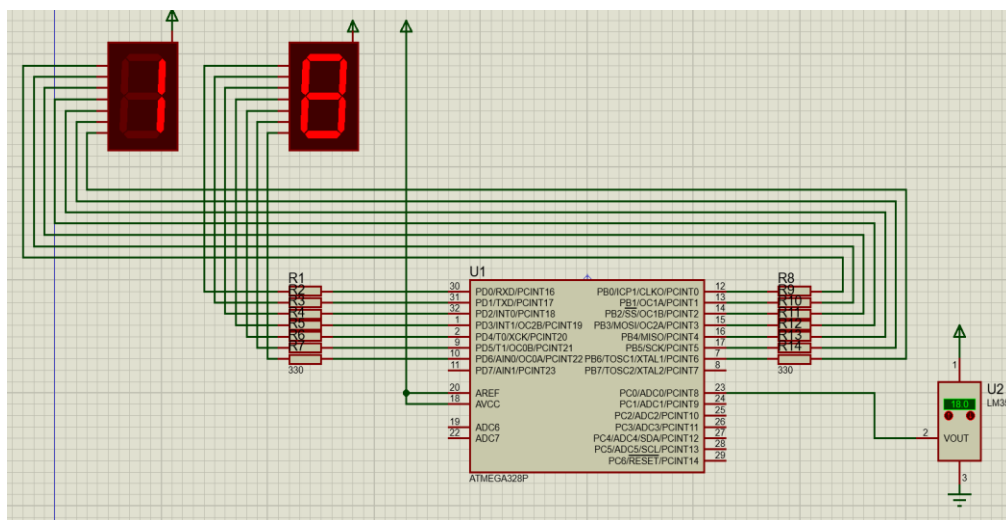
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INTRODUCTION

The purpose of Practicum 1 was to design and test an embedded system that measures temperature using an LM35 temperature sensor and displays the temperature value on two 7-segment displays. The practicum was first simulated in Proteus using an ATmega328P microcontroller. The measured temperature had to be updated immediately and remain accurate within approximately 5%.

DESIGN

- Components Used.
 - Atmega328P Microcontroller.
 - LM35 Temperature sensor.
 - BCD Seven-segment Display.
 - Resistors.
- Proteus Circuit design



The circuit was designed in Proteus using the Atmega328P, LM35 temperature sensor, and two 7-segment displays. The LM35 sensor output pin was connected to one of the microcontroller ADC input pins. The Controller continuously read the analog voltage from the sensor and converted it into a temperature value. The temperature value was separated into tens and units' digits. These digits were then sent to the corresponding 7-segment displays.

- Display 1 outputs the tens digits.
- Display 2 outputs the units digits.

Current-limiting resistors were connected between the microcontroller outputs and the display segments to protect the LEDs.

- Program Code

```
#define F_CPU 1000000UL
```

```
#include <avr/io.h>
```

```
#include <util/delay.h>
```

```

const uint8_t digits[10] =
{
    0b01000000, // 0
    0b01111001, // 1
    0b00100100, // 2
    0b00110000, // 3
    0b00011001, // 4
    0b00010010, // 5
    0b00000010, // 6
    0b01111000, // 7
    0b00000000, // 8
    0b00010000 // 9
};

void adc_init(void)
{
    ADMUX = (1 << REFS0); // AVCC reference, ADC0

    ADCSRA = (1 << ADEN) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0); //
prescaler 128
}

uint16_t adc_read(void)
{
    ADCSRA |= (1 << ADSC);

    while (ADCSRA & (1 << ADSC))
    {
    }

    return ADC;
}

uint8_t read_temperature(void)
{
    unsigned int adc_value = adc_read();

    // LM35: 10mV per degree
    // With 5V reference: temp = ADC * 500 / 1024
    unsigned int temp = (adc_value * 500UL) / 1024;

    if (temp > 99)
    {
        temp = 99;
    }

    return temp;
}

int main(void)
{
    DDRB = 0xFF; // display 1
    DDRD = 0xFF; // display 2
    DDRC = 0x00;

    adc_init();

    while (1)
    {
        unsigned int temperature = read_temperature();
    }
}

```

```

        PORTB = digits[temperature / 10];
        PORTD = digits[temperature % 10];

        _delay_ms(200);
    }
}

```

CHALLENGES AND SOLUTIONS

- Unstable temperature reading – check ADC configuration.
- Compilation errors in the program – corrected syntax and verified byte data.
- Difficulty understanding ADC conversion – reviewed the AVR procedures on ADC.

RECOMMENDATIONS

- Ensure that all wiring connections are checked carefully before running the simulation.
- Use proper resistor values to protect the 7-segment displays.
- Use short and neat wiring connections in Proteus to improve readability.
- Test individual sections of the circuit separately before combining the full system.

CONCLUSION.

In this practicum, a temperature monitoring system was successfully designed and simulated using Proteus software. The system used an Atmega328P, an LM35 temperature sensor, and two 7-segment displays.

The Atmega continuously reads the analog voltage from the LM35 sensor and converts it into a digital temperature value using the ADC. The temperature value was then displayed correctly on the two 7-segment displays.

The practicum improved understanding of analog-to-digital conversion, temperature sensing, display interfacing, and Proteus simulation. Several challenges were experienced during the design and testing process, including incorrect display outputs and unstable readings, but these were resolved through troubleshooting and verification of the circuit connections.

Overall, the practicum was successful and demonstrated the practical application of embedded systems concepts in temperature measurement and display systems.